# ActInf Textbook Group ~ Cohort 1 ~ Meeting 16 (Chapter 6, part 2)

*TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 16 (Chapter 6, part 2) | 54:55*

Okay, hello, it's September 16th, we're in Cohort 1 of the Active Textbook Group, it's meeting 15, or I guess, yes, no, 16, on, let me just update the title, meeting 16, we're in our second discussion on Chapter 6.
So, are there any overall comments that people would like to address about Chapter 6?
Or any other questions or points to add, or just areas of Chapter 6 that they found to be interesting?
Distinction between the interceptive-extraceptive modalities, which have been mentioned in Box 6.1. And first, I think, Pristen, with Michael Allen, if I'm not mistaken, has published a recent paper just today, exactly on that topic, in which they've somehow tried to provide a computational model for the first time, for the unification of the interceptive and extraceptive modalities. So, since this chapter is also about the modeling of those agents, I think this paper from the, in the body's eye, the computational anatomy of interceptive inference might be relevant for anyone who wants to model these modalities, with the aid of active inference framework.
Thank you. Very nice. Thanks for sharing it.
Let's look at Box 6.1.
As people see it, what distinguishes exteroceptive, proprioceptive, and interoceptive modalities?
For example, why are proprioceptive and interoceptive separated?
Because I couldn't hear you. I don't know if it's on my side or connection was unstable from your side. Sorry.
Let me try something. I'll be gone for a second.
Am I back?
Okay. So, yeah. What? Yeah, Brock, I do need the star link.
Why are proprioception and interoception separated?
And why are these three categories modeled as different?
Why have they been framed as different?

Why do they be framed as different from our body?
Why do they be framed as different from our body?
Why do they be framed as different from our body?

biological and neurological mechanisms and neural mechanisms that somehow can control those
different kind of perceptions because for the extra septic perception the organism would
necessarily need a kind of sensory I mean the sensation it gets from the environment is just
on the level of its of the imminent stimuli that it gets from the external environment but for the
interceptive modality the sensation is just the internal sensations and so on and for the
pre-preceptive it's related to motor control feedback loop so I guess because of their different
mechanisms it might be helpful to talk in different
terms about each of them but I'm not totally sure about that
I agree I think one aspect even in a pre-blanket or non-blanket modality is like they're about
external things versus about internal things and then between proprioception and interoception
you're absolutely right that they have different neural mechanisms and physiological mechanisms
um with some blurry space perhaps for example like interoception about blood pressure
is somewhere in between like a muscular reflex and uh uh an organ reflex but the tissues being addressed
are different um let's look at this paper again we'll we'll return to the book but we're going to be
thinking today and then especially going forward with like how do you identify the um let's go to this
question which we just just to look at it which system are we modeling what is the appropriate form
for the generative model how do we set up a generative model how do we set up a generative process
so let's see if we can determine some of those aspects in this paper
hashtag figure 4-3 it is the very same
okay this is just a general pointing to a partially observable Markov decision process
with with with a slight um plus computational biology graphical veneer to match the color scheme
probably not but but it kind of looks like it
here is where we see the schematic
of the form of the generative model
variable. Variables are used to reflect that those variables can be fit from empirical data
or just a priori set and simulated, but this is the form of the model. These are the shapes of
the matrices and tensors that go from, that are going to address what their model is about.
Does anybody want to like describe what they see in this?
What do you think they were trying to explain?
And what do you think they were trying to model? What is their system of interest based

upon what you see here?
Actually, one interesting term that I came up with, I mean, used in this paper is
cardiac active inference, the term that they've used in this paper to describe the kind of
interrelationship between the heart rhythm and the, I mean, the interception. So this generative model,
I believe, is just about modeling that relationship between the heart rhythm and the interception. So
we can see, as usual, the B matrices, C and A matrices that we've seen before. And, but in this case,
they've been applied to the variation in the heart rhythm and how it relates to the interception.
And so, yes. So again, using the variables that we know from figure four, three in the book,
and their figure one here, the hidden states that are not directly observed are on the interoceptive
side relating to the state of the heart. On the interoceptive side, whether there is a flower or a spider.
The outcomes that are observed are the visual interoception of a flower or a spider and the heart rate,
probably a low and a high heart rate. The piece that links these two is the sensory attenuation
between the high heart rate state and the mapping of a steroception.
In the B matrix, you can just see that even the allocation of the symbols is different.
Like when you're aroused, when you're in the, this, this is like heart open and then heart contract.
So when you're in the aroused situation, B is how the hidden states change their time.
You're always going to go from here to here.
Whereas in the relaxed situation, sometimes you go to the, from this state to this one.
Other times you're relaxed. And so you actually, you dwell within that state.
So here it's like pump rest, pump rest, pump rest here. It's 50% slower. Just, it depends on the
parameters, but it's 50% slower because sometimes you're going to go from one, but it looks like as
it's here that it's actually like, or it's twice as slow. Rest, rest, beat, rest, rest, beat.
So these are the two interoceptive states.
One has a preference for having relaxed hearts when flowers are around and having contracted
heart when spiders are around. The A matrix maps on these two sensory modalities between
hidden states and outcome states, just like any A matrix would do in any POMDP.
And then there's also a transition matrix of the generative process.
So it's a very simple generative process where either flowers are flowers or spiders are spiders.
As they would say, you can use standard routines to simulate these techniques. I hope that the
code are available. You can plot parameter sweeps as you explore the beta and the alpha.
Yeah. So here's like the beta parameter and alpha parameter. So you can do parameter sweeps with
this grid search or other approaches and then do statistics on those outputs. Is this quote active
inference? Well, it's the output of a parameter sweep on an active inference model. There's various
other kinds of methods. So not to spend too, too much time on the paper, but does anyone have any
thoughts on like that case or how they see it applying to chapter six's content?
I think it speaks to a fulfillment cycle in active inference research.
Oh,
yeah.

yeah.
This work example helps a lot.
I mean, if anyone wants to,
especially because describing this kind of cardiac active inference as a kind of system that has a definite Markov blanket, it was really interesting question at the first place. And I think they've done a really great job, obviously, in that regard, and thinking really creatively and out of the box about how we can describe that whole system with native using Markov blankets to describe its internal and external environment and so on.
It also relates to this earlier question. So how does the figure four, three paradigm discrete and continuous state spaces with all the attending complexities? Do you hear me, Ali?
Okay.
Yeah.
Okay. Okay. How does this representation
connect to?
Can you hear Daniel?

I can.
It's a story for another day. Some of the incredible,
the truly incredible internet.
I'm going to try one thing. So I think that
or Ali, can you hear me now or no?
Yeah. Yeah.

something

here's what's interesting
with the wi-fi
and then we'll continue
because it's not it's not
a wi-fi learning group
but with the hotspot
there are a set of people
who i cannot interact with
but with wi-fi i am able
to interact with those
people
just a note so
we see the pomdp representation
now what is not in this paper
here is a graphical layout
of the pomdp
okay
what is not in this paper
a figure 6-1
or indeed a figure 1-1
representation
so why is that
i'm not sure i've understood your question
quite clearly but
could you please elaborate on that a little bit more
yes
yes sure
so when we look at papers like
vassal et al
communication is active inference
social and active active inference
loops
loops
loops
you know
vatkin et al
remote teams
loops

loops
loops
markov blanket
interface
loops
entities
things
yet in the application
what is specified
is not
the loopy
situation
what needs to be specified
is chapter 6
not chapter 1
so i find that very interesting
because so much of the
focus
attention
discourse
has been around
abstractions around
markov blankets
and their interpretations
from various stances
and about
boundaries
physical boundaries
and conceptual boundaries
but it's a non
controversy
when it's written this way
and
there are markov blankets
between many of these variables
because it's a bayesian graph
that's not fully connected
so every bayesian graph

that's not fully connected
has markov blankets
trivially
but
this
paper
doesn't
need to
describe
what is
this is enough to say
what's inside and outside
the organismal
markov blanket
if we want to even think about it
is like
here
in this line
kind of
not
not
on the bayes graph
but it's between
the spider and the flower
outside
and the vision
of the spider and the flower
but that's like
taken care of
with the A matrix
of the
exteroceptive modality
which is

I think
why this
recipe
does not

focus
on
which
thing
are we
modeling
but rather
which system
are we
modeling
and yes
we're going to
pursue the generative
model before the
generative process
even though they may
have symmetry
or they may be very
similar
but I just
find it very
interesting
open to
hearing what people
think
like
why
the
cybernetic
loop
action
perception
loop
is used
so
much
in contrast
with the

form
of the
model that's
actually
implemented
which is
it's a
different
topology
and a
different
discussion
I know
that they're
integrated
and
compatible
just one
comment on
that is
you see
at least
to my
understanding
Markov
blanket
is
a
kind
of
a
kind
of
think
modeling
tool
or
a

kind
of
way
of
thinking
about
the
basically
the
ontology
of
I mean
ontology
in the sense
of
philosophical
use
the
ontology
of the
system
that
we're
about
to
model
so
it's
not
necessarily
the
relationships
among
the
different
components
of the
Markov

blanket
is
something
not
necessarily
physical
or
tangible
relationships
in that
system
so
it
might
be
kind
of
just
informational
as
we've
just
we saw
in
I think
chapter 5
if I'm not
mistaken
the
information
passing
among the
different
components
and so
on
so
at least

I think
the reason
for not
using
the
schematic
model
of
Markov
I mean
the
cybernetic
model
of
Markov
blanket
instead
of
using
just
the
schematic
form
probably
it's
because
of
this
kind
of
non
let's
say
non
physical
or
non
tangible

relationship
among
the
different
components
by
non
tangible
and
non
physical
I mean
something
that
can't
be
explained
in
terms
of
its
ontological
existence
if
I don't
know
if I
could
express
myself
in that
regard
or not
but
thank you
Ali
Brock
and anyone

else
I was
just
going
to
I don't
know
I
feel
like
personally
I'm
just
kind
of
at
my
affordances
that I
have
to
get
pragmatic
and
epistemic
value
here
without
doing
some
modeling
of some
kind
yeah
I don't
know
that's
just

good
thanks
I mean
this is
the
second
discussion
on
chapter
six
and
then
this is
setting
us up
to
actually
pursue
a
model
so we
should
be
providing
these
things
yeah
I'm
just
yeah
I'm
just
saying
it's
definitely
ripe
for
that

I
mean
yes
Rohan
yeah
so
actually
just to
get some
clarification
you're
asking
why
the
form
of
the
model
is
as
mentioned
in
chapter
six
of
topology
given
in
chapter
six
which
is
chapter
one
yes
I mean
they're
they're

compatible
topologies
that actually
are describing
an overlapping
set
of
notes
I was
just
calling
attention
to
the
amount
of
discourse
and
focus
on the
potentially
more
familiar
more
schematic
action
perception
loop
and the
way that
that draws
so much
attention
to the
organism
environment
interface
whereas

when it
comes to
the
empirical
modeling
that
interface
never had
to be
specified
as like
a first
class
type
it just
was included
by the
by
we just
said there
are spiders
and flowers
out there
hidden
states
s
and then
there's
observations
of
and there
there's
no
error
there's
no
error
right

but
there
would
be
there
would
be
no
error
in
the
generative
process
itself
the
model
might
have
error
right
so
um
so
there
would
be
the
that
comes
from
some
measurement
issue
right
so
so
when
you

measure
it
and
you
confuse
say
spider
for flower
or flower
for spider
then
you would
get some
error
and there
may be
that's where
I guess
you can
talk about
the marker
blanket
uh
mispecifying
or
misobserving
something
so
I
I'm
not
sure
what
exactly
is
the
I
mean

they're
compatible
right
so
in
this case
there's
no
error
but
if
there
was
error
obviously
there
would
be
something
wrong
let me
actually
just
correct
this
I
believe
be
I
didn't
mean
to
say
this
is
the
visual
error

this
is
saying
that
they're
unchanging
flowers
so
it's
either
they're
either
one
or
the
other
here
parametrically
it's
possible
to
introduce
visual
error
in
the
A
matrix
but
I'm
just
pointing
to
the
architecture
of
the
model

doesn't
require
even
engaging
with
this
it's
totally
compatible
and
the
observations
are the
sense
states
and
the
actions
are the
active
states
and
so
on
but
as
presented
and
as
constructed
we
if we're
going to
use a
POMDP
architecture
we don't
necessarily

even need
a representation
like this
yeah
Jessica
okay
that's
interesting
hi
yes
for me
the reason
why
like
it's
like
two
part
like
the
reason
why
the
reason
craft
is
so
popular
and
I
think
maybe
it's
also
the
reason
why
it

was
used
and
this
is
the
simplicity
to
show
the
concept
and
the
relationship
so
like
the
new
one
it
great
for
I guess
for
the
paper
that
it
was
created
but
it
doesn't
show
the
relationship
as
clear

as
the
other
ones
it's
very
simple
so
I
think
that's
why
it
was
used
maybe
like
remove
a lot
of
extra
information
that
the
second
paper
illustrate
complexity
in
it
but
the
simplicity
of
that
graph
is
why

it
really
encapsules
the
concept
and
it's
also
the
reason
why
it's
so
popular
because
it's
so
simple
I
think
that's
the
relationship
from
how I
see it
yep
totally
makes
sense
thank
you
and
once
we
see
that
there

is
a
mapping
from
the
schematic
cybernetic
view
then
we
can
probably
fill
in
a lot
of
the
parts
of
the
PMDP
but
just
to
look
one
more
level
here
I
mean
the
code
I
posted
the
code
below

so
it's
in
MATLAB
which
people
may or
may not
have
access
to
oh
sorry
okay
Ali
and
then
Rowan
no
I
just
wanted
to
point
out
one
thing
that
I
forgot
to
mention
before
one
thing
that
might
be

helpful
to
keep
in
mind
is
that
actually
in
this
generative
model
there
is
a
kind
of
unification
or
kind
of
integration
of
interceptive
and
exerceptive
modalities
because
as
they
pointed
out
in
the
paper
this
agent
that

they're
modeling
adopt
two
modes
of
engagement
with
the
world
one
of
which
is
the
relaxed
and
the
other
one
is
the
aroused
mode
of
engagement
and
because
of that
there
are
two
sorts
of
hidden
states
related
to

each
interceptive
and
extra
of
one
so
I
guess
if
they
wanted
to
put
that
kind
of
interrelationships
between
these
kind
of
modalities
in
that
simplified
schematic
model
it
wouldn't
necessarily
clarify
these
kind
of
integration
between
the

modalities
so
one
reason
might
be
related
to
that
peculiar
situation
of
this
particular
case
here
thanks
it
also
speaks
to
the
step
in
the
model
of
the
form
of
the
generative
model
and
then
I
think
when

people
get
married
to
the
cybernetic
scheme
you
can
imagine
well
then
there's
two
nodes
inside
of
the
internal
states
there's
two
modes

you
can
go
between
but
now
you're
mixing
and
matching
what
arrows
mean
where

some
of
the
arrows
are
reflecting
causal
influence
and
some
of
them
are
reflecting
transitions
between
mutually
exclusive
states
and
so
then
we
end
up
back
into
the
wireframe
nightmare
where
it's
like a
schematic
where
some
edges
are

tactile
connections
and
some
are
causal
links
and
some
are
mutually
exclusive
transfers
of
information
and
all
these
other
ways
that
people
visualize
models
and
this
representation
reminds
us
that
we're
dealing
with
variables
and
states
and
connections

Jessica
I
think
that
was
from
before
but
yeah
but I
guess
I
think
it's
like
the
visual
it
has
to
do
with
what
you're
trying
to
convey
in
that
moment
and
you
know
how
much
depth
of
information

do
you
need
I
think
like
at least
say
like
for
this
one
you
need
extra
information
so
the
other
model
wouldn't
work
well
just
share
like
with
the
idea
of
what
the
model
is
maybe
this
version
with

the
extra
information
might
be
too
complex
so
the
other
one
there's
a
better
purpose
so
it's
like
what
is
the
purpose
you
trying
to
accomplish
and
what
do
you
want
to
communicate
and
depending
on
that
then

models
with
more
information
or
less
information
are
more
susceptible
yep
thank
you
totally
agree
I mean
they
didn't
do
any
of
the
loop
at
all
they
just
went
straight
to
the
POMDP
representation
and also
this isn't
the only
form
of an

interoceptive
exteroceptive
integration
model
one could
brainstorm
a whole
host
and again
that's
what the
recipe
is for
well
we could
have
a
mutually
exclusive
transfer
between
these
two
states
defined
by
these
two
B
matrices
which
entails
a
policy
variable
switching
between
the

two
or
we
could
have
a
continuous
variable
called
arousal
and
arousal
is engaged
in some
sort of
attenuation
relationship
here
so
this isn't
even the
only
form
but
it
shows
us
a
form
okay
soon
in
20
minutes
we're
going to
be heading
into

chapter
7
so
let's
preview
it
so
that
when
we're
working
on this
in the
next
week
preparing
our
questions
starting
to think
about
what
we want
to be
modeling
we'll
kind of
know
what's
going to
happen
all right
chapter
7
is
active
inference
in

discrete
time
people can
still
hear me
okay
I think
so
so
here
is where
we
cross
the
rubicon
with
modeling
this is
abstract
up till
now
you cut out
there for a
second
oh okay
okay
it's all
good now
yeah
just
okay
thank you
chapter 7
we're crossing
the rubicon
together
and
it's been

abstract
up till
now
first
what is
addressed
is
perceptual
processing
you'll find
a lot
of resonance
between
this
chapter 7
and
step by
step
which is
model stream
one
maybe it's
step hyphen
yeah
yeah
this paper
step by
step tutorial
is going
to do
a certain
walkthrough
of
adding
incremental
complexity
into the
model

in terms
of model
parameter
depth
okay
it's
model
stream
one
there's
four
streams
that
Ryan
and
Chris
White
did
static
perception
dynamic
perception
dynamic
perception
with
preferences
and free
energy
minimization
dynamic
perception
with free
energy
minimization
and
flexible
policy
selection

through
uncertainty
modulation
we're going
to do
something
similar
in
chapter
seven
they
don't
do
that
upper
left
corner
of
step
by
step
they
go
straight
to
hidden
Markov
models
for
dynamic
perception
this is
analogous
to a
Coleman
filter
or any
other number

of
general
Bayesian
filtering
schemes
we have
a
prior
D
hidden
states
S
and a
sequence
of
observables
through
time
T
that are
mapped
to
hidden
states
A
B
describes
how
hidden
states
change
through
time
so we
have
the
temperature
is a

hidden
variable
and then
we're getting
observations
from the
thermometer
that are
mapped to
the real
temperature
by A
maybe it's
a good
thermometer
maybe it's
biased
or whatever
and then
there's the
temperature
changing
through
time
then
there's a
very
Ali
centric
discussion
of listening
to music
this has
to be
one of
the most
glaring
errors

in the
textbook
it's
very
Warhol
like
though
so maybe
it has
a certain
aesthetic
to it
but the
example
that's
going to
be
provided
is
like
a
musical
listening
example
so
the
A
matrix
is
emitting
notes
based
upon
the
hidden
state
which
is

the
score
of
the
music
the
B
matrix
is
and
they
have
70%
accuracy
so
this
one-tenth
distributes
through
the
matrix
so
it's
just
to
use
integers
here
but
really
this
is
0.7
0.1
0.1
0.1
because
it

has
to
add
up
to
1
but
when
this
person
when
the
hidden
state
of
the
score
is
in
the
first
note
they
play
it
70%
of
the
time
and
then
the
other
three
notes
they
make
an

error
10%
of
the
time
each
time
70%
of
the
time
the
musician
hits
their
note
the
B
matrix
is
how
things
change
through
time
the
hidden
state
so
this
is
the
transitions
in
the
score
and
here

there's
a
97%
probability
again
it's
distributed
throughout
with this
one over
100
just to
represent it
with integers
and
there's a
prior with
no uncertainty
or ambiguity
of just
it begins
in the
first
position
and so
this
we can
even see
like
do
do
do
do
you know
I don't
want to
use
ABCD

but it
would be
like
BCDA
in the
musical
ontology
so here's
BCDA
like that
but then
there is
an error
in this
one simulation
run
it'd be
probabilistic
but they're
showing for
didactic
reasons
a run
where the
third
point
in the
third
observation
there is
an error
however
it's heard
as the
correct
value
that's
a pure

inference
example
this is
not
active
inference
this is
just
Bayesian
models
of
perception
this
kind of
sessile
creature
is
rather
uninteresting
okay
now we
move from
a hidden
Markov
model
crotching
tiger
hidden
Markov
into
a
partially
observable
Markov
decision
process
a hidden
Markov

model
is a
partially
observable
Markov
model
because
some parts
of it
are hidden
some are
not
but the
key piece
here is
the introduction
of the
decision
we know
that we
model that
with policy
that influences
how states
change through
time
and that
policy
selection
is governed
by g
figure
4-3
people can
still hear
because I
see all
these like

frozen for
me
okay
all right
okay thank
you yeah
it's just
strange
stuff
here's
4-3
again
here
when we're
talking about
policy
selection
we're talking
about
states that
haven't
happened
yet
observations
and hidden
states that
haven't
happened
yet
we see
a restatement
of equation
2.6
with expected
free energy
here broken
into epistemic
and pragmatic

value
now
to flesh
out
this
POMDP
we turn
to a
classic
Par
and Friston
example
which is
the
T-Mace
we're
not going
to go
through
every
single
part
of this
T-Mace
right now
but
it'll be
a key
area
for us
to unpack
why
do the
matrices
have the
dimensions
that they
have

where are
the
four locations
that the
mouse can
be
here
down at the
bottom
to the left
or to the
right
so this
is not
like a
mouse
moving
in
continuous
space
it's like
there are
four locations
it can be
in
those are
the
columns
there's
five
hidden
states
of
location
which
we can
see
are like

based
upon
whether
when it
goes
down
to the
bottom
to get
a
queue
whether
the
hidden
state
is going
to reveal
the queue
or not
and then
figure
seven
four
and
seven
five
you'll
see
differ
first off
one of
them is
bolded
this one
is
bolded
this one
is

unbolded
but
here
we see
a one
in the
L
column
because
the
black
is
on
the
L
and
here
we
see
a
one
in
the
R
column
because
the
black
is
on
the
R
so
understanding
why
that
is
is

going
to be
the
crux
of
understanding
how
to
construct
an
epistemic
informative
queue
as a
hidden
environmental
variable
in a
generative
model
here
are
the
B
matrices
transition
probabilities
here's
the
preference
vector
we'll
talk
about
what
C1
and
C2

are
here
are
the
priors
we'll
talk
about
what
D1
and
D2
are
now
there's
more
unpacking
of the
epistemic
value
itself
so
in
the
equation
above
there
was a
restatement
of
equation
2.6
which
partitions
the
expected
free
energy

into
the
negative
epistemic
value
and the
pragmatic
value
now
we're
going to
follow
up
on
the
epistemic
value
I
of
pi
the
epistemic
value
of
a
policy
and
break
that
down
into
these
representations
here
are
simulations
simulated
epistemic

and pragmatic
behavior
of a
rat
foraging
in a
t-mace
so
we'll
try to
model
what
these
runs
are
I
think
it
might
be
one
small
note
about
the
book
that
especially
if
they
knew
that
it
was
going
to
be
black

and
white
which
it
seems
like
that
was
known
this
is
not
ultra
helpful
perhaps
people
disagree
but
another
this
is
what
I said
another
warhol
yeah
this
one
this
one
you
can
even
see
the
Campbell
soup
but

this
I
think
if
someone
were
familiar
with
reading
these
traces
they
would
still
want
to
understand
what
these
different
lines
were
and
like
what
the
dashed
one
is
but
if
people
are
unfamiliar
reading
these
kinds
of

error
residual
traces
!
I
don't
know
what
is
being
shown
here
figure
7.7
that
was
the
simulation
of
the
rat
getting
the
epistemic
cue
and
going
to
the
left
so
instead
of
just
taking
a
50-50
shot

on
where
the
food
was
the
rat
navigates
this
explore
exploit
dilemma
it
engages
in
an
epistemically
oriented
action
or an
action
driven
by
epistemic
value
to
reduce
uncertainty
get
the
cue
and
then
it
secures
the
food
in

this
box
7.1
we
see
a
discussion
about
parameterization
of
uncertainty
and
how
precision
is the
inverse
of
variance
and
this
omega
is
the
precision
variable
here
just
like
Mark
Solms
uses
but
others
use
different
ones
the
lower

this
value
is
the
more
even
things
are
and
so
this
has
come up
in
the
ice cream
example
with
the
folk
psychology
live
stream
like
if
your
c
vector
is
0
6
negative
6
versus
0
500
negative
500

that's
like a
sharper
preference
distribution
so
it's
the
quantitative
values
in
the
c
for
example
that
specified
not just
the
direction
but also
the
intensity
of
preference
and
then
analogously
precision
can be
specified
in a
continuous
form
here
is a
reference
to

earlier
Par and
Friston
work
on
vision
that
has to
do
with
ocular
motor
foraging
which
is
heavily
epistemically
driven
but
then
they
are
using
this
example
to
explore
how
precision
the
viewer
shows
a
version
to
the
upper
left

square
when
it
is
specified
with
a
less
precise
more
ambiguous
likelihood
mapping
the
lower
left
square
is
epistemically
attractive
when
the
transition
probabilities
are
specified
as
more
uncertain
we'll
try to
understand
these
examples
learning
and
novelty
7.5

here
we
take
that
POMDP
and
like
we
push
it
back
in
the
genealogy
now
we
have
a
priors
on
the
hidden
states
which
are
themselves
equipped
with
prior
beliefs
these
are
Dirichlet
conjugate
priors
but
they
can

be
of
other
forms
but
they're
just
shown
as
Dirichlet
here
because
it's
like a
convenient
statistical
distribution
family
for
variational
inference
which
we
explored
in
several
chapters
ago
these
can be
considered
as
hyper
priors
because
they
but
in the

end
just
variables
on
a
graph
and
it's
not
like
they're
tagged
with
being
a
hyper
prior
and
you
can
imagine
priors
that
exploring
these
potentially
open-ended
Bayesian
graph
structures
is why
it's
so
important
to have
structure
learning
like

Carl
has
pointed
out
for
years
and
why
we
want
to
have
parameter
sweeps
across
cognitive
model
structures
in
active
block
fronts
because
why
stop
why
not
have
another
prior
on
this
one
so
we
should
have
the

notation
and
the
capacity
to
sweep
that
here
is a
discussion
on
conjugate
priors
a
conjugate
prior
means
that
when
used
to
perform
Bayesian
inference
the
posterior
belief
will be
the same
type
of
distribution
so
previously
we talked
about
how
the

hidden
state
could be
a
continuous
variable
and
the
internal
state
could be
a discrete
variable
you know
the
temperature
is
continuous
outside
and
then
inside
is it
hot
or
is it
cold
is
like
what
I'm
caring
about
so
the
families
don't
have

to
match
but
when
we
use
conjugate
priors
it
makes
it
so
that
where
you
can
design
it
to
play
nice
it
does
play
nice
but
we'll
work
through
that
also
with
anybody
who
wants
to go
into
more

detail
on
the
math
because
these
are
just
like
Bayesian
statistics
points
here
theta
is
used
to
just
describe
parameters
it's
a
very
common
notation
but
not
100%
pervasive
to
use
theta
for
parameters
just
generally
here
there's

more
unpacking
of the
update
rules
this
is
probably
a
secondary
consideration
for
just
constructing
a
model
here
we
return
to
expected
free
energy
and
several
more
representations
beyond
what was
seen
above
and
beyond
what was
seen
in
equation
2.6

here
the
learning
model
that was
presented
above
the
POMDP
with
another
generation
of
genealogy
is
described
which is
drawn
from
these
two
papers
then
structure
learning
is
introduced
we'll
see how
deep
this
one
goes
but
these
papers
are
quite

interesting
and
connected
to
Bayesian
model
reduction
here
a
hierarchical
model
is
presented
here's
figure
4-3
in the
dashed
box
now
there's
a
figure
4-3
you know
yo dog
I heard
you liked
figure
4-3
then
there's
another
hierarchical
model
simulation
from the
above

showing
belief
updating
again
with
these
trace
diagrams
and
then
there's
a
summary
so
it's
a
somewhat
long
chapter
it
contains
like
the
music
example
the
rat
example
the
maze
example
the
eye
example
so
we're
seeing
like

data
points
in
model
space
in
the
last
three
minutes
does
anyone
have
any
thoughts
or
questions
how
are
they
going
to
tackle
chapter
seven
and
as
they
start
to
construct
their
model
if
they
want
to
I

am
just
every
direction
that I
can
because
it is
quite
dense
still
and
kind of
unapproachable
I think
for me
anyways
it just
is quite
difficult
to get
a grasp
on it
from
the
textbook
specifically
at this
point
like the
generative
model
what does
that mean
in
code
and where
does that

correlate
to this
particular
function
or that
particular
time
step
or
whatever
it's
yeah
I don't
know
it's
just
a lot
and so
I'm
just
trying
to
from
yeah
I don't
know
from
just
generally
building up
simulations
which are
just things
that you
can
you know
put together
in a

browser
like
with
simple
JavaScript
to like
CAD CAD
and
Active
Block
Rents
and
these
sorts
of
things
and
trying
to
grasp
at some
real world
situation
so maybe
that
that
rat
you know
T maze
is a
really good
place to
focus on
for me
like a
good
intersection
between the

book
and
something
that
isn't
like
the frog
jumping
or is it
not
jumping
is
you know
so contrived
it's really
difficult
to
think about
modeling
that
and
in a
way
that would
help
explain
what's
going
it's
like
yes
I knew
the frog
was jumping
or not
jumping
so that
doesn't really

help
right
whereas
the rat
maybe
coming up
with some
left or right
hidden state
based on the
queue
that's partially
there's
there's
some
explanatory
power there
that may
be realized
I think
so
I'm just
yeah
I don't
know
it's
I'm
trying to
bridge
jump over
the grand
canyon
of the
philosophical
conceptual
okay got
it
but

the math
there's a
okay
but there's
a gap
there for
me
so
I'm trying
to pull
them
together
that's
yeah
I have
a question
related to
calibration
of those
priors
and
things
is that
ever
explained
as to
how
to go
about
doing
that
for
any
of
these
models
yes
so

model-based
data
analysis
is going
to describe
okay
so
it's
okay
it's
in the
future
all right
chapter
seven
describes
just in
our
last
minute
chapter
seven
describes
the
discrete
state
models
chapter
eight
describes
continuous
time
models
chapter
nine
describes
using
empirical

data
for
model
driven
data
analysis
chapter
10
is a
summary
the
textbook
is
you know
yeah
we're
excited
towards
the
grand
canyon
but I
think
again
we have
several
months
to
unpack
this
together
so
I
hope
that
we
can
go

quite
far
with
!
so
please
stay
engaged
and
add
your
questions
and
start
building
from
both
sides
play
with
the
code
examples
provided
break
them
and
then
start
from the
other
side
with
a
recipe
on
a
situation

that

you

care

about

and

then

they're

going to

meet

in the

middle

okay

thank

thank you

everyone

farewell

bye

everyone

bye

bye

bye